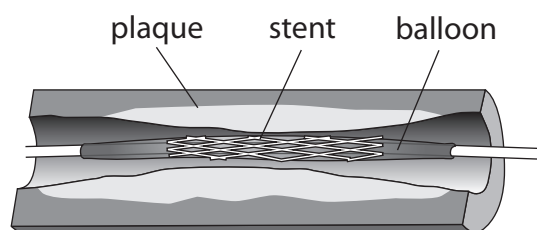
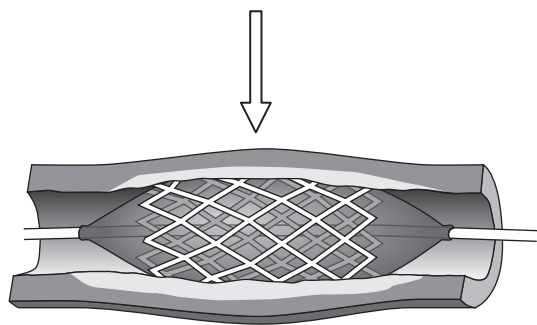


### 3 Stents are used in the treatment of atherosclerosis.

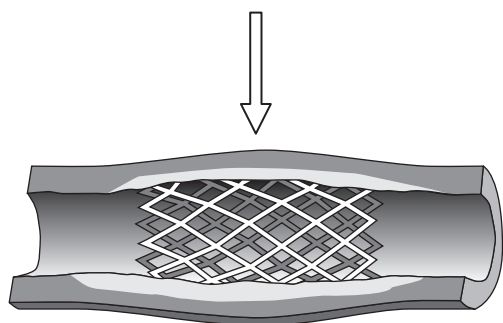
(a) The diagram shows how stents are positioned in a diseased coronary artery.



An uninflated balloon and narrow stent are inserted into the diseased coronary artery.



The balloon is inflated, expanding the stent.



The balloon is deflated and removed, leaving the expanded stent in place.

Explain why a stent is used in the treatment of atherosclerosis in a coronary artery.

(3)

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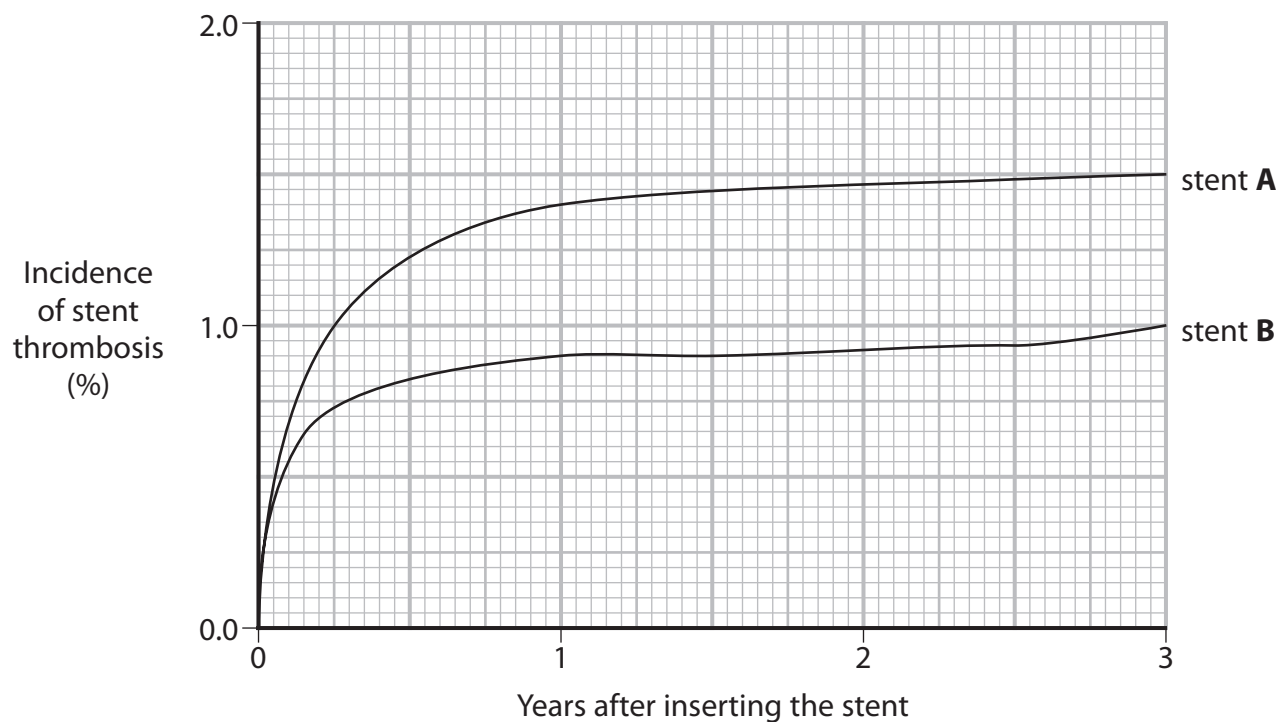


(b) Inserting a stent can damage the artery.

This damage can result in stent thrombosis if the blood clotting process is stimulated.

A study looked at the damage caused by two different types of stent, stent **A** and stent **B**.

The graph shows the incidence of stent thrombosis found in this study.



- (i) In this study, 800 patients had stent **A** inserted and 400 patients had stent **B** inserted.

Calculate the difference in the number of patients who developed stent thrombosis three years after inserting the stent.

(2)

Answer .....



- (ii) State how the formation of a blood clot in the coronary artery could cause the death of the patient.

(1)

- (iii) Which components of the blood clotting process are enzymes in their active form?

(1)

- ☐ A prothrombin and thromboplastin only
- ☐ B prothrombin and thrombin only
- ☐ C prothrombin, thrombin and thromboplastin
- ☐ D thrombin and thromboplastin only

- (iv) Which components of the blood clotting process are soluble in blood plasma?

(1)

- ☐ A fibrin and fibrinogen
- ☐ B fibrin and thromboplastin
- ☐ C fibrinogen and thromboplastin
- ☐ D fibrinogen only

- (v) Stent **B** contains a drug to prevent stent thrombosis.

Suggest one type of drug that could be used in stent **B**.

(1)

(Total for Question 3 = 9 marks)

